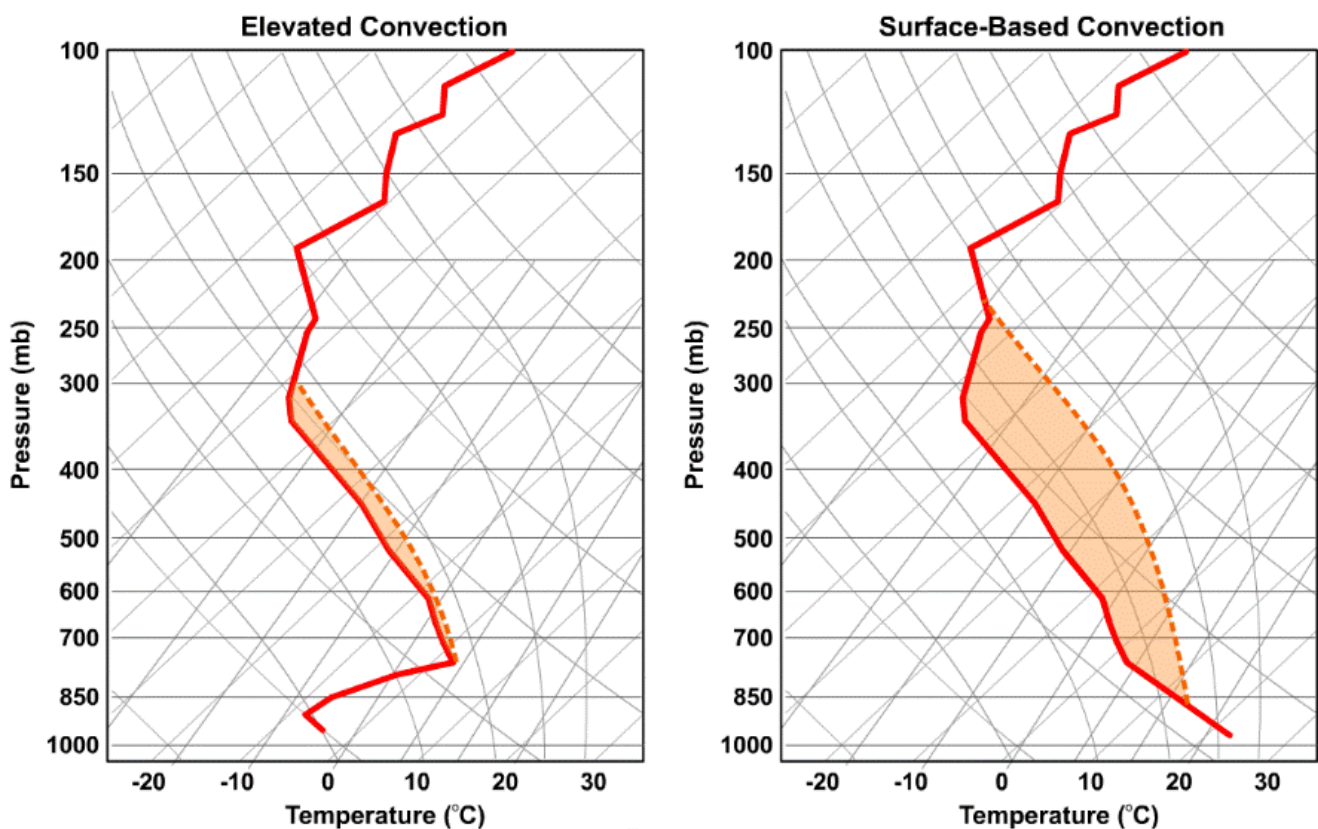


ELEVATED CONVECTION

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To get a better idea about the contrast between elevated thunderstorms and surface-based thunderstorms, check out the idealized skew-T diagrams below. The sounding on the left is consistent with elevated convection. Note the stable layer (a layer of relatively warm air) between 900 mb and 750 mb (roughly). If unstable parcels of air are lifted from the top of this stable layer, they become positively buoyant, setting the stage for elevated convection (although CAPE is rather small). The sounding on the right favors surface-based convection (surface air parcels lifted to the LFC become dramatically positively buoyant through a deep layer).



(Left) If unstable parcels of air are lifted from the top of the stable layer between 900 mb and 750 mb (approximately), they become positively buoyant, setting the stage for elevated convection (even though CAPE is rather small). (Right) This sounding favors surface-based convection (surface air parcels lifted to the LFC become dramatically positively buoyant through a deep layer). Note that CAPE is noticeably larger.

Credit: David Babb

So, forecasters look for pockets of warm advection at 850-mb because they can signal overrunning. If lapse rates above the cool, stable layer at the surface are sufficiently steep, elevated thunderstorms can develop, especially if some upper-level divergence is present to give parcels an additional kick of upward motion to get things started.